



Design of slippery organogel layer with room-temperature self-healing property for marine anti-fouling application

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ABSTRACT

Slippery liquid-infused porous surfaces (SLIPS) was proven to be a potential strategy for inhibiting marine biofouling, which is an economic and environmental problem. The traditional SLIPS fabricated based on capillary effect of micro-structure was prone to be damaged in practical application. To solve this problem, a slippery organogel layer (OG) was designed by infusing silicone oil into poly(dimethylsiloxane) based polyurea (PDMS-PUa) synthesized from α , ω -Aminopropyl terminated poly(dimethylsiloxane) (APT-PDMS) and isophorone diisocyanate (IPDI). The self-healing property of the substrate was attributed to the breaking and reforming of hydrogen bonding in urea groups between PDMS-PUa, and self-generating of the lubricant oil layer was based on infiltration of silicone oil through PDMS-PUa substrate to the damaged area. The as-fabricated OG exhibited slippery performance with a low water sliding angle of $< 10^\circ$. The fractured substrates could self-heal after contacting at room temperature for 48 h, and the self-healing efficiency of the substrates increased with temperature. In addition, the as-prepared OG exhibited excellent performance in inhibiting bacterial settlement compared with bare glass (BG) and PDMS-PUa verified by bacterial settlement experiment under both static and dynamic environments. The as-prepared OG will provide an alternative strategy to design self-healing marine anti-fouling coatings in practical application.

1. Introduction

Biofouling is associated with various fields including but not limited to biosensors, aquaculture industry, heat exchangers and maritime industry [1–4]. In many cases, marine biofouling, the accumulation and settlement of organic molecules, microorganisms and macrofoulers on submerged surface in the sea water, was the complex and intractable world-wide problem [5,6]. The process of marine biofouling was initiated by the formation of a conditioning film due to rapid accumulation of organic molecules, which was followed by the settlement of microorganisms forming biofilm and subsequent macrofoulers such as sponges and barnacles [6–8]. And the marine fouling had adverse effects on the service life as well as operational cost of marine equipments, environmental problem and so forth [6,9,10]. To solve these problems, it was necessary and important to explore effective marine anti-fouling methods.

The earliest marine anti-fouling method, the use of toxic heavy metal antifoulants such as lead, mercury and relevant organic

derivatives, was banned for causing serious environmental pollution [7,11]. Then the biocide tributyltin (TBT) and other organotin compounds containing anti-fouling coatings were widely used for its excellent capability to deter marine organisms for many years [11,12]. However, the release of organotin compounds had adverse effects on other marine animals [13]. For this reason, the organotin compounds containing anti-fouling coatings were ultimately banned by International Maritime Organization (IMO) in 2008 [5,12]. After that, the biocide-free compounds containing fouling release coatings (FRCs) were increasingly used for reducing adhesion strength of fouling on the surfaces based on its low surface energy [7,8]. But the removal of the fouling was relied on hydrodynamic shear force generated by submerged surface movement or dynamic water movement, which was unfavorable to deter fouling under static conditions. Recently, bio-inspired antifouling strategies, mimicking the chemical composition, physical structure and anti-fouling mechanism of creatures, have attracted increased attention for its environmentally-friendly and excellent anti-fouling performance in different fields [6,14–22].

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Pitcher plants, a kind of carnivorous plants, were able to capture preys based on slippery liquid film locked in the porous microstructured peristome [17,23]. Inspired by pitcher plants, the slippery lubricant-infused porous surface (SLIPS) fabricated by infusing perfluorinated lubricant liquids into porous teflon membranes was first reported by Aizenberg's group [17]. The as-prepared SLIPS presented liquid-repellency, low contact angle hysteresis, self-healing, as well as stability under water and high pressure [17,24–28]. It has reported that the SLIPS could effectively inhibit common bacteria such as *P. aeruginosa*, *S. aureus*, and *E. coli* settlement on the surfaces [29]. After that, intense studies gradually found SLIPS were capable of inhibiting marine bacteria and macrofouling organisms such as *Chlorella vulgaris*, sulfate reducing bacteria (SRB), *Desulfovibrio* sp., zoospores, cyprids, mussel and so on settlement on surfaces owing to intact lubricant oil layer [21,24,26,29–32]. To improve the stability of slippery surface under severe conditions, Wong et al. and other research groups fabricated slippery materials with self-healing substrate for anti-fouling and inhibiting organic liquids [17,33,34]. Howell et al. designed slippery anti-fouling surfaces with self-healing lubricant oil layer for improving its stability [35–37]. Besides, to our knowledge, researchers rarely designed self-healing slippery materials with self-healing lubricant oil layer and self-healing substrate after huge mechanical damage in anti-fouling field [38]. However, the continuous lubricant layer of traditional SLIPS, depended on the capillary force, was prone to be destroyed after severe mechanical damage of the porous substrate in practical application. It can be inferred that designing stable slippery surface can be achieved from two aspects: self-healing lubricant oil layer and self-healing substrate.

On one hand, inspired by secretion of epicuticular wax on the surface of land plant leaves [39], a self-healing lubricant layer can be designed to inhibit fouling. For example, Wang, L. et al. fabricated a solid organogel materials with secretion of alkanes by swelling PDMS with long chain alkanes [40]. On the other hand, to improve the stability of organogel, the self-healing solid substrate was indispensable in practical application. To surmount this, reversible associations such as disulfide bond, hydrogen bonding, dynamic-covalent boronic esters, metal-ligand coordination and dynamic imine bond could be incorporated to fabricate intrinsic self-healing substrates [41–47]. For example, Xu, J. et al. fabricated a poly(urea-urethane) (PUU) based composite materials with intrinsic room-temperature self-healing capability through hydrogen bonding in PUU matrix [47]. Therefore, combining regenerable liquid layer of organogel and self-healing substrate, a self-healing organogel can be designed to inhibit marine bio-fouling in practical application.

In this work, a room-temperature self-healing slippery OG was fabricated by infusing the silicone oil into poly(dimethylsiloxane) based polyurea (PDMS-PUa) synthesized from α , ω -Aminopropyl terminated poly(dimethylsiloxane) (APT-PDMS) and isophorone diisocyanate (IPDI). The as-fabricated OG surface presented liquid-repellency performance verified by the sliding velocity measurement with a low tilted angle of $< 10^\circ$. And the healed sample could sustain the external force and exhibited liquid-repellency after self-healing at room temperature for 48 h. The self-healing efficiency of fractured substrate increased with temperature. Furthermore, under both static and dynamic environments, the OG surface performed excellent anti-fouling performance compared with the bare glass (BG) and PDMS-PUa after immersion of 7 and 14 days in the culture solution inoculated with *Pseudoalteromonas* sp. Our study demonstrated both the lubricant oil layer and substrate of OG were able to self-heal at room temperature, which will provide a novel method to fabricate self-healing marine anti-fouling coatings in practical application.

2. Materials and methods

2.1. Chemicals and materials

Slide glasses were supplied by Yancheng Feizhou Bosu Co., Ltd. China. α , ω -Aminopropyl terminated poly(dimethylsiloxane) (APT-PDMS, $M_w = 1000$) was purchased from Hangzhou Silong Mater. Tech. Co., Ltd and dried under vacuum at 120°C to remove the water and impurities before use. Silicone oil (PDMS, 350 cst) was purchased from Dow Corning. Isophorone diisocyanate (IPDI, 98%) was obtained from Sigma-Aldrich. Tetrahydrofuran (THF), ethanol and acetone were analytically pure and purchased from Sinopharm Chemical Reagent Co., Ltd. THF was distilled at 75°C to remove the water before use. Distilled water was used throughout the experiment.

2.2. Preparation of organogel layer

IPDI (2.2284 g) in THF (12 ml) and silicone oil (15 g) were successively added into a three-necked flask. Then APT-PDMS (10.00 g) in THF (100 ml) was dropped into flask for 1 h in nitrogen environment. Finally, the mixture was stirred at room temperature for another 3 h and was monitored in situ by Fourier Transform Infrared Spectrometer (FTIR) (Thermo Electron Corporation, Nicolet iS10, Washington, USA) in nitrogen environment. Once the isocyanate band at 2260 cm^{-1} was absent entirely, the product was poured into mold and cured in vacuum oven at room temperature for 48 h.

2.3. Self-healing of substrate

The as-prepared sample was cut into two pieces and then put into contact without extrusion. The self-healing process conducted without external force at different temperatures and was recorded by digital photograph.

The maximum tensile force (F) of sample to be sustained was tested by a hand compression tensile tester (ALIYIQI, AST-S, Yueqing, China) with a digital tension indicator (100 N) three times. The maximum tensile stress of the sample (σ) was calculated by:

$$\sigma = F/S$$

Here, F was the maximum tensile force, S was the cross section area of the sample, σ was the maximum tensile stress of the sample.

2.4. Sliding velocity measurement

The water droplet slid on the as-prepared and healed OG surfaces with a low tilted angle of $< 10^\circ$, which was measured by the contact angle meter (Powereach, JC2000C1, Shanghai, China). The sliding distance of water droplet was determined by extracting one image every 30 s.

2.5. Bacterial settlement analysis

The bacteria (*Pseudoalteromonas* sp.) samples obtained from Bohai Sea were used as model bacteria for the bacterial settlement experiments. And the culture solution was obtained by adding bacteria into the culture medium, which was prepared by charging the yeast extract (1.00 g), peptone (5.00 g) and Fe_2SO_4 (0.01 g) into sea water (1 L) [36]. After sterilization by UV radiation for 30 min, the three types of samples with the hot glue could be adhered closely to the beaker (250 ml) before charging culture solution (200 ml) into it for the bacterial settlement experiments under both static and dynamic environments at 30°C . And

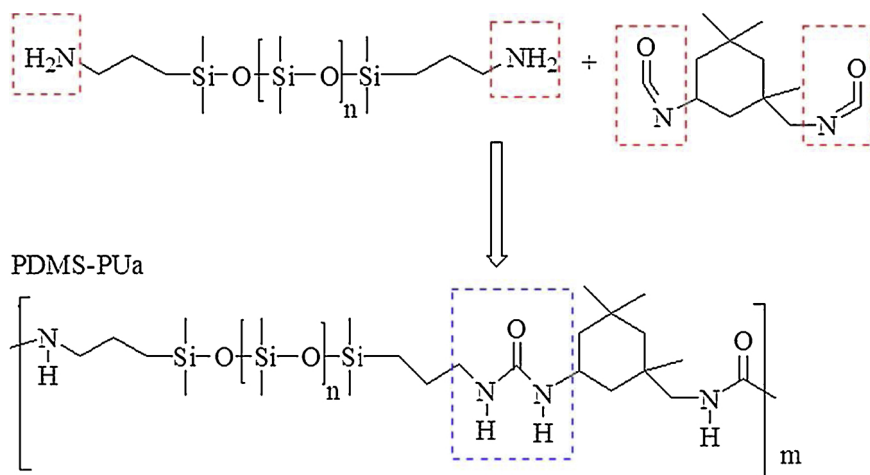


Fig. 1. Schematic illustration of synthesis of PDMS-PUa.

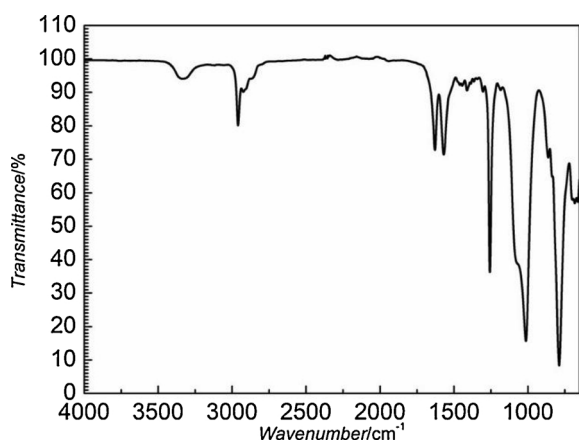


Fig. 2. FTIR spectrum for PDMS-PUa.

the settlement experiment under dynamic environment was conducted on the samples with seawater flow speed of ~ 0.4 m/s, which was achieved by shaking the orbital shaker with speed of 120 rpm. The seawater flow speed was set as ~ 0.4 m/s, which is close to mean tidal speed in Bohai Sea [48]. By adding fresh culture solutions to take place of same volume of the culture solutions in the settlement experiments every 3 days, the bacteria kept active under both static and dynamic environments.

To determine the settlement of bacteria on the samples, the fluorescence microscope (ZEISS, Axio Imager A2, Jena, Germany) was utilized to observe the bacteria stained by the DAPI solution with concentration of 100 ng/mL for 15 min on the samples after immersion in the culture solution of 7 and 14 days [36]. After that, fluorescence images can be used to determine the bacteria coverages on the samples with the software Image J.

3. Results and discussion

3.1. Composition of the PDMS-PUa based slippery OG

Fig. 1 presented the synthesis of PDMS-PUa polymer, which was synthesized from α , ω -Aminopropyl terminated poly(dimethylsiloxane) (APT-PDMS) and isophorone diisocyanate (IPDI) based on the reaction of amine group and isocyanate. Then the PDMS-PUa based slippery OG was fabricated by infusing silicone oil into the reaction system of α , ω -Aminopropyl terminated poly(dimethylsiloxane) (APT-PDMS) and isophorone diisocyanate (IPDI). The end of the reaction was confirmed by the disappearance of the isocyanate band at 2260 cm^{-1} in the Fourier Transform Infrared Spectroscopy (FTIR) (Fig. 2). The PDMS, a component of the organogel, played a vital role in fabricating room-temperature self-healing anti-fouling organogel. On one hand, PDMS with a low glass-transition temperature endows the PDMS-PUa based polymer high chain mobility, which is favourable to the reforming of the weak hydrogen bonding existing in the urea groups at room temperature [49–51]. On the other hand, PDMS has a low surface free energy and good compatibility with various organic liquids including silicone oil [51,52]. Therefore, PDMS based materials should be promising candidates for fabricating self-healing slippery surface.

3.2. Wettability of organogel layer

Fig. 3 presented the dynamic motion of water droplet on the surface of as-fabricated OG surface. As shown in Fig. 3, the water droplet could slide on the as-fabricated surface with a low tilted angle of $< 10^\circ$, which was because the water was immiscible to lubricant oil on the surface. Furthermore, the result indicated the continuous and chemical homogeneously lubricant layer formed on the surface, otherwise the water droplet could hardly slide on the as-prepared slippery surface.

3.3. Self-healing property

To examine the self-healing property of substrate, the sample was cut into two pieces and put into contact without any extrusion for the self-healing in air at room temperature (Fig. 4A). After contacting for

Fig. 3. Dynamic mobility of water droplet on the as-fabricated OG surface with a low tilted angle of $< 10^\circ$.

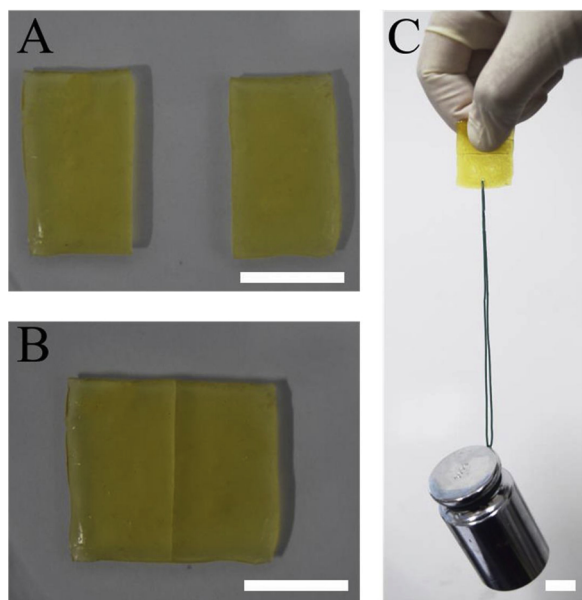


Fig. 4. The self-healing process of OG at room temperature (scale bar = 1 cm): A damaged sample, B healed sample, C healed sample can withstand the external force (the mass of weight = 500 g).

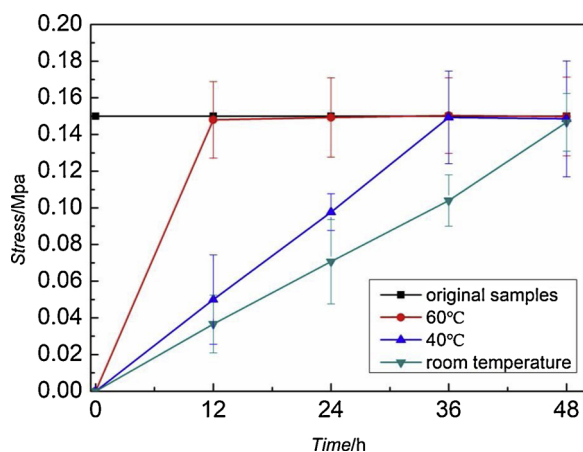


Fig. 5. The maximum tensile stress of the healed OG at different temperature as a function of time.

48 h, the two pieces of the sample autonomously healed (Fig. 4B), and the healed sample could sustain the external force (the mass of weight = 500 g) without any deformation due to the breaking and reforming of hydrogen bonding at the damaged interface at room temperature (Fig. 4C). Hence, the results demonstrated that the as-prepared OG surface exhibited self-healing capability based on breaking and reforming of hydrogen bonding in the urea groups at room temperature.

The self-healing efficiency of fractured sample, calculated by the ratio of $\sigma/\sigma_0 \times 100\%$ (σ_0 , maximum tensile stress of the original sample), was also quantified by tensile tests at different temperature

conditions. As shown in Fig. 5, it was obvious that the self-healing efficiency of fractured sample increased with the self-healing time. Besides, when the self-healing efficiency reached 100%, i.e., the maximum tensile stress of the healed OG reached to that of original OG, the self-healing time of healed sample was 48 h at room temperature. While at 40 °C and 60 °C, the self-healing time decreased to 36 h and 24 h respectively, suggesting the self-healing efficiency of fractured sample also increased with the temperature. It was because the breaking and reforming of the heat-labile hydrogen bonding was accelerated at a higher temperature, thus decreasing the self-healing time [50].

To examine the self-healing capability of the lubricant oil layer, we further studied the dynamic motion of water droplet on the healed OG surface. As shown in Fig. 6, after self-healing at room temperature for 48 h, water droplet still could slide easily on the healed OG surface with a low tilted angle of $< 10^\circ$. It indicated the continuous and chemical homogeneously lubricant layer reformed on the healed OG surface due to infiltration of the silicone oil through substrate to the damaged area.

3.4. Anti-fouling performance

In this bacterial settlement experiment, the partial samples were took out every 7 days and stained to observe the coverages of *Pseudoalteromonas* sp. on the surfaces through fluorescence microscope. As shown in Figs. 7 and 8, it was clear that *Pseudoalteromonas* sp. on BG and PDMS-PUa surfaces were much more than those on as-prepared OG surface under both static and dynamic environments. Furthermore, to quantify the bacterial coverages on the surfaces, the statistical *Pseudoalteromonas* sp. coverages on the surfaces were summarized in Fig. 9 based on the fluorescence microscope images in Figs. 7 and 8. The concrete statistical *Pseudoalteromonas* sp. coverages on the surfaces were also listed in Tables 1 and 2. As shown in Table 1, after immersion of 7 days under static environment, the statistical bacterial coverages on the BG, PDMS-PUa and OG were $\sim 22.5\%$, $\sim 20\%$ and $\sim 0.1\%$ respectively. The results indicated the noninfused PDMS-PUa could hardly inhibit bacterial settlement on surfaces compared with OG. However, the statistical bacterial coverages on OG showed a decrease of 2 orders of magnitude compared with those on BG and PDMS-PUa, indicating the as-prepared OG surfaces performed excellent anti-fouling performance due to the existence of lubricant oil layer on the surface. It has been considered the main anti-bacteria mechanism of SLIPS was the existence of liquid-like lubricant oil layer, where the bacteria hardly anchor under static environment [21,29]. In addition, after immersion of 14 days under static environment, the statistical bacterial coverages on the BG, PDMS-PUa and OG were $\sim 29\%$, $\sim 27\%$ and $\sim 0.3\%$ respectively. In comparison with 7 days, the statistical bacterial coverages on OG showed a slight increase ($\sim 0.2\%$) compared with those on BG ($\sim 7\%$) and PDMS-PUa ($\sim 7\%$), suggesting the OG was stable and able to effectively inhibit bacterial settlement on the surfaces after immersion of 14 days. So it demonstrated the as-prepared OG had excellent anti-fouling performance under static environment, which was mainly due to the existence of the liquid lubricant oil layer on the OG. Numerous studies have reported the biofouling coverages were great reduced on the SLIPS due to the existence of lubricant oil layer on it [21,24,26,29,53]. It was known the surface topography, roughness and other surface properties had influence on the biofouling coverages on the surface [21,54–56]. The OG with chemical homogeneously silicone oil layer was liquid-like

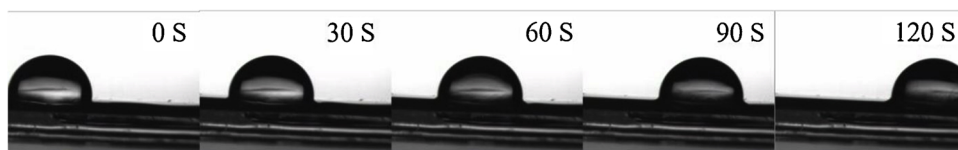


Fig. 6. Dynamic motion of water droplet on the healed OG surface with a low tilted angle of $< 10^\circ$.

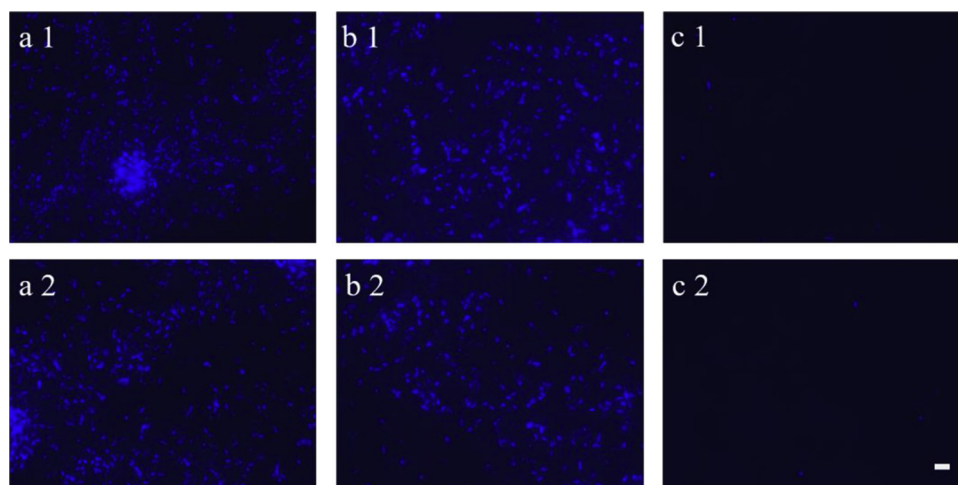


Fig. 7. The fluorescence microscope images of samples after immersion in static sea water 1 and dynamic sea water 2 inoculated with *Pseudoalteromonas* sp. for 7 days; a Bare glass (BG), b PDMS-PUa, c OG.

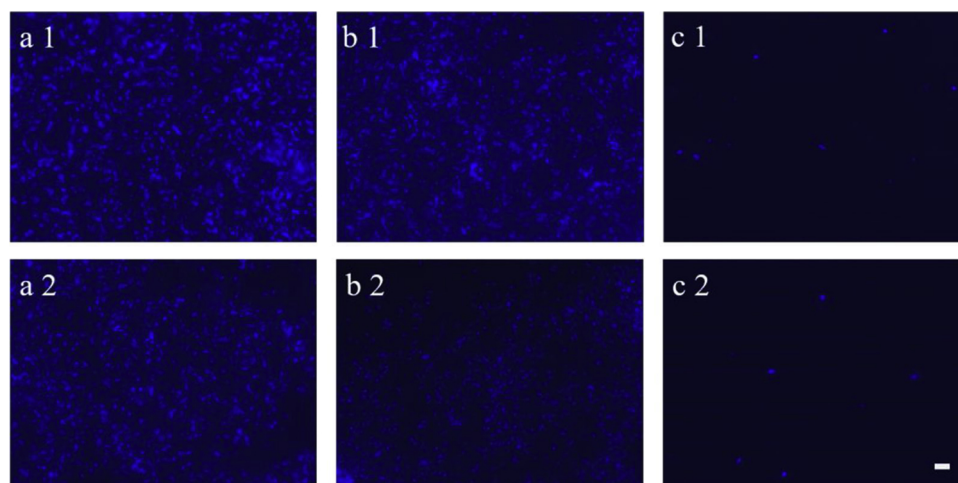


Fig. 8. The fluorescence microscope images of samples after immersion in static sea water 1 and dynamic sea water 2 inoculated with *Pseudoalteromonas* sp. for 14 days; a Bare glass (BG), b PDMS-PUa, c OG.

surface with low roughness, which was facilitated to inhibit bacterial settlement on the surface. The main reason was that the liquid-like property of lubricant oil layer was barrier for bacteria to anchor and attach to the surface [21,29], which could effectively inhibit bacterial settlement to the OG surface.

The anti-fouling performance and stability of the as-prepared OG are important features in marine environment. We further conducted the bacterial settlement experiments under the simulated dynamic environment. After immersion of 7 days under dynamic environment, the statistical bacterial coverages on the BG, PDMS-PUa and OG were ~20%, ~18% and ~0.05% respectively. Similar to the results under static environment, the OG exhibited excellent anti-fouling performance compared with BG and PDMS-PUa under dynamic environment. However, in comparison with the results under static environment, the statistical bacterial coverages on the three types of surfaces decreased in a certain extent under dynamic environment. The main reason was that the bacteria on the surfaces were partially washed away by the shear force generated by the dynamic water. In addition, compared with 7 days, the statistical bacterial coverages on OG immersed for 14 days were ~0.2%, indicating the lubricant oil layer on the as-prepared OG was still stable under dynamic environment. However, the lubricant oil layer may be destroyed for the longer immersion time. It is interesting and meaningful to conduct further research on improving the stability of as-

fabricated OG surfaces for longer immersion time.

4. Conclusion

In summary, we fabricated a self-healing slippery OG by infusing the silicone oil into PDMS-PUa with dynamic hydrogen bonding. The as-prepared OG presented liquid-repellency performance with a low tilted angle of $< 10^\circ$. And the healed OG presented liquid-repellency performance owing to the self-healing of substrate and self-generating of lubricant oil layer. The self-healing of substrate was based on the breaking and reforming of hydrogen bonding between PDMS-PUa at room temperature. The self-healing efficiency of fractured substrates increased with temperature. And the lubricant oil layer was capable of self-generating through infiltration of lubricant oil to damaged area. Furthermore, under both static and dynamic environments, the OG layer performed excellent anti-fouling performance compared with the BG and PDMS-PUa after immersion in the culture solution inoculated with bacteria. Besides, in comparison with static environment, the bacterial settlement on the surfaces were lower under dynamic environment, which was attributed to the shear force. Therefore, the as-prepared room-temperature self-healing OG had the potential to the practical application in the field of the marine anti-fouling.

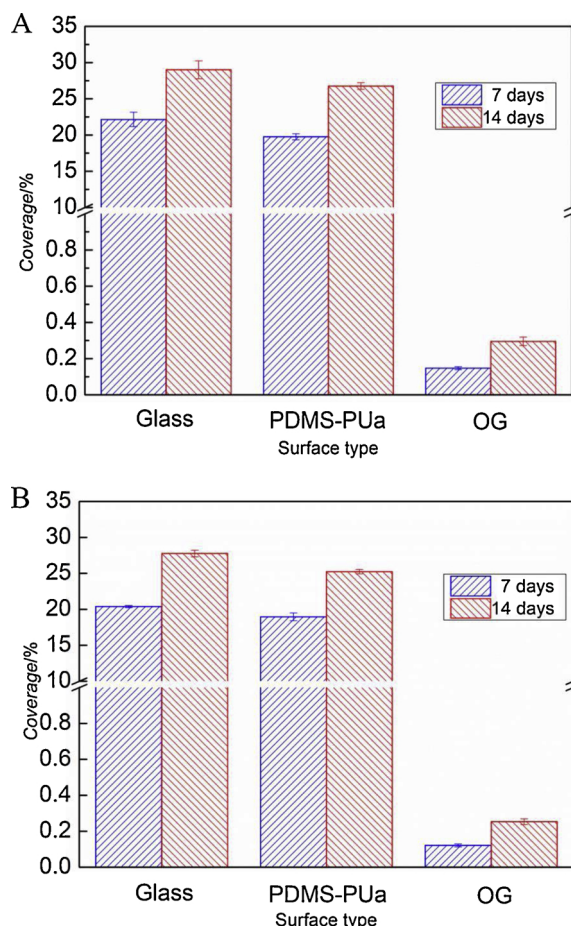


Fig. 9. Statistical results of bacterial coverages on samples (BG, PDMS-PUa, OG) after immersion in static (A) and dynamic (B) culture solution inoculated with *Pseudoalteromonas* sp. for 7 and 14 days.

Table 1

Statistical results of bacterial coverages on surfaces (BG, PDMS-PUa, OG) after immersion in static culture solution inoculated with *Pseudoalteromonas* sp. for 7 and 14 days.

Surface type	Glass	PDMS-Pua	OG
7 days	~22.5%	~20%	~0.1%
14 days	~29%	~27%	~0.3%

Table 2

Statistical results of bacterial coverages on surfaces (BG, PDMS-PUa, OG) after immersion in dynamic culture solution inoculated with *Pseudoalteromonas* sp. for 7 and 14 days.

Surface type	Glass	PDMS-Pua	OG
7 days	~20%	~18%	~0.05%
14 days	~27.5%	~25%	~0.2%

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